SOFTWARE REQUIREMENTS SPECIFICATION FOR TM VISUALS

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1. Project Introduction

This is a photography and client management project for TM Visuals that consists of a website and mobile app connected to the same database. The project aims to ease customer interaction and promote the organisation’s exposure on the digital platform.

2. Stakeholder Introduction

2.1 StakeHolder Profile and Organisation

(Overview)

Name	Tshedza Mulaudzi
Role	Sole Proprietor of TM Visuals
Organisation Name	TM Visuals
Type of Organisation	Creative agency in the visual arts and creative industry

Organisation's Main activities	Provide professional photography and videography services Serves: students, individuals, event organisers, restaurants brands and businesses Attract new clients through social media platforms. Showcase previous work to potential clients. Make it easier for clients to contact TM Visuals. Deliver completed photographs and videos in high quality via google drive to protect private client content. Maintain professional communication with clients. Keep the business name and online identity consistent as TM Visuals.
Contact Person	The primary contact person for the project is Tshedza Mulaudzi, the owner and primary operator of TM Visuals. He is responsible for photography, videography, client communication, quotations, booking management, editing, file organisation and delivery of completed client work.
Reason for selecting this stakeholder	TM Visuals was selected as the stakeholder because it has a real and identifiable business problem rather than a hypothetical one. The business currently does not have a centralised website or booking system and relies mainly on social media platforms for bookings, communication, and content delivery. The requirements of the business align well with the technologies chosen for the project, namely Flutter, ASP.NET, and Supabase. Additionally, the stakeholder meeting provided clear and detailed information about booking processes, pricing, deposits, content delivery, and privacy concerns. This gives the group a realistic and well-defined project scope that can be completed within the allocated project timeline.

2.2 Current Process and Problem Experienced

Current Process

A potential client discovers TM Visuals through social media or a word of mouth referral. They contact the photographer directly through WhatsApp or TikTok to enquire about services. The client and photographer then discuss the service required, the date, and availability through messaging. Client and owner negotiate if necessary. A deposit amount which is 50% of the commission needs to be paid before any work will be carried out. This is to avoid last minute cancellations. Once a date is agreed, the booking is confirmed manually

and is only recorded in chat history or the photographer's personal notes. On the agreed date, the photography or videography session takes place. Afterwards, the photographer edits the photos and videos, uploads the high-resolution files to Google Drive, and sends the client a shareable link so they can download their media. Files are organised using clients name, date and content type.

Who participates

- Client
- TM Visuals (Photographer)

Information usage

- Chat messages through social media (WhatsApp or TikTok) containing enquiry details, dates and price discussions.
- TM Visuals's personal notes of which dates are already booked.
- Google Drive link used to deliver the final files.

Current-process diagram:

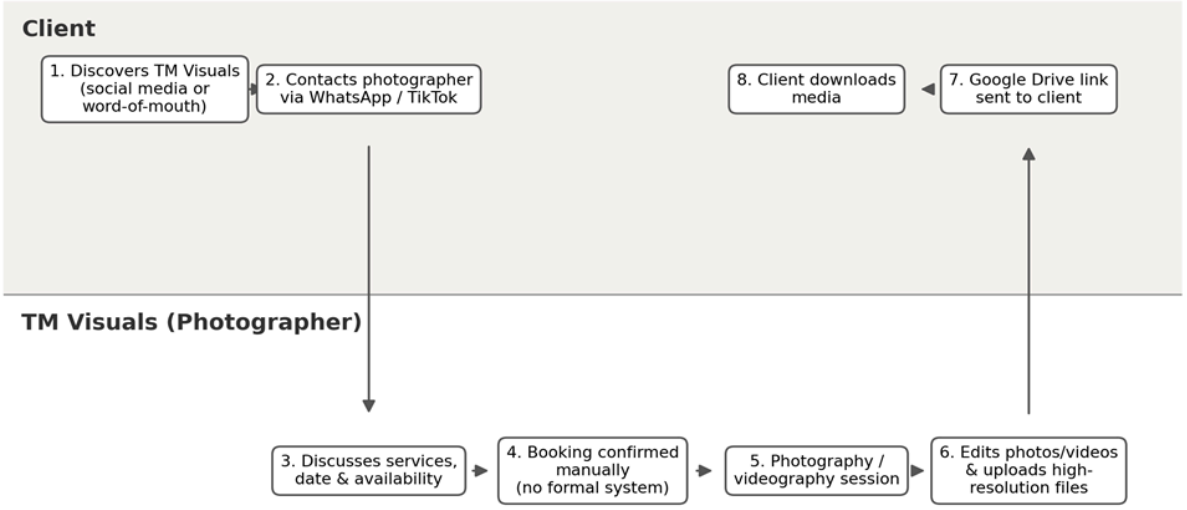


Figure 1: Current process - TM Visuals and Client booking and delivery flow

2.3 Problem Experienced

Main problem

The current way of managing the business creates several challenges. Since most communication takes place through social media, potential clients do not have one place where they can easily view the services offered, see examples of previous work, and find important business information. Bookings are handled through these social media platforms. Important information can become difficult to track. Availability has to be managed manually and this leads to potential risks of double-booking or conflicting bookings. TM Visual's exposure relies heavily on word-of-mouth which is outdated in a digital era.

How it affects the organisation and its users
Social media bookings are going to cause dual booking resulting in scheduling conflicts. Another problem is if photographs from previous shoots are shared with potential clients, the privacy of those clients needs to be considered. Users who require TM Visual's services have a difficult time finding and requesting their services.

3. Problem Statement

TM Visuals currently operates using only social media platforms which make it difficult to expose their services and display previous works done. Their method of exposure is via word-of-mouth which makes it difficult to garner more clients.

This creates difficulties for potential clients who cannot access social media platforms. It increases the administrative effort involved in managing bookings and availability, creates a risk of double-booking, and makes high-quality client content delivery less efficient. TM Visuals therefore requires a centralised digital platform that provides an accessible portfolio, structured booking and availability management, clear booking policies and high-quality client content delivery.

3.1 Current Situation & Consequences

Current situation	Mr. Mulaudzi currently relies on Instagram, TikTok, and WhatsApp for client communication and business promotion. The business does not have a centralised website or booking system. Bookings, price negotiations, and content delivery are managed manually through direct messaging and Google Drive. Bookings are scheduled manually.
Affected users	- Individuals, - Students, - Event Hosts/Organisers, - Brands and Businesses, - Restaurants and other prospective customers - Clients who are not on social media platforms - Mr Mulaudzi (business owner)

Consequences	The lack of a centralised system may lead to scheduling conflicts and double bookings. Clients who do not use social media have limited access to the portfolio, reducing opportunities to attract new customers. Communication across multiple platforms is difficult to manage, and limited online visibility may restrict business growth. Important information can become difficult to track if they use a manual booking process.

4. Project Aim & Objectives

4.1 Overall Purpose Of The Project

Build a system that integrates high quality content delivery, secure portfolio access, and real-time booking tracking. Improve client access to services, portfolio, bookings and completed work. Provide secure and organised delivery of client content. Reduce manual administration and booking conflicts.

4.2 Main Objectives to be achieved

Provide a centralised and Professional presence for TM Visuals.
Allow clients to check available dates.

- Prevent double-booking.
- Allow clients to submit booking requests.
- Allow the owner to approve or manage bookings.
- Allow clients to track booking status.
- Protect private client photographs and videos.
- Provide secure access to completed client content.
- Maintain flexible, customised pricing.
- Provide clear Terms and Conditions.
- Improve communication between TM Visuals and clients.

4.3 Stakeholder and User Requirements

Stakeholder requirements:

- A booking system that allows users to book for certain days and be able to see days that are booked.
- The stakeholder should be able to manage bookings and enquiries.

- Pricing should not be the same for all services but should be customized.
- Personal client photos or videos should be protected from unauthorised access.
- Be able to upload their profile to showcase work.
- Must have an introduction/overview of the stakeholder and what they do.
- Non-refundable deposit must be explicitly stated under services offered.
- The download feature should only be available for the user's own content.
- Categorise portfolio content
- Manage private client galleries and access permissions.
- Send secure links for completed client content.
- Receive notifications for new enquiries/bookings.
- Keep organised records of clients, bookings and completed work.
- Manage booking information when arrangements change.
- Link the website to Instagram, TikTok and WhatsApp.

User Requirements:

- Users must be able to contact the owner.
- Users will be able to access their content exclusively.
- Users must be able to make a booking and submit enquiries.
- Must be able to view the stakeholder's introduction/overview.
- Users can cancel a booking with acknowledgement of the non-refundable booking fee.
- Users must be able to view the photography and videography services.
- Users must be able to view available and unavailable booking dates.
- Users must be able to see the status of their booking request.
- Browse portfolio content by category.
- Users can view frequently asked questions.
- Access their completed content through secure access.
- Download their permitted content in high quality.
- Access TM Visuals' social-media platforms.
- Users can reschedule their booking.
- Use the system from mobile and desktop devices.

5. Project Scope

5.1 Scope Overview

The project aims to develop a professional and responsive website and mobile app for TM Visuals to improve its online presence, increase its visibility to potential clients, and provide a more convenient way for clients to interact with the business. The system will allow potential clients to learn more about the business, the owner, view its photography and videography portfolio, submit enquiries, and request bookings. It will also assist the business owner with managing bookings, availability, website content, and client information. The system will provide existing clients with secure access to their completed content while supporting high-quality media

delivery. Overall, the project aims to centralise important business activities and provide a more organised and professional experience for both TM Visuals and its clients.

5.2 Detailed Scope Components

Project Title	TM Visuals
Project Aim	Develop a professional website and mobile app prototype connected to the same database for TM Visuals Centralise important business activities in one platform. Give potential clients an easy way to: - Learn about the business and its owner, - View previous work. - Submit bookings. - Negotiate prices. Help the owner manage bookings, availability and website content. Give existing clients secure access to their completed content.
Objectives	Establish a professional online presence, provide an accessible and categorised portfolio, streamline client enquiries and bookings, improve availability management, prevent double-booking, support customised quotations and deposits, protect private client content, improve high-quality media delivery and provide the owner with suitable tools for managing websites content and bookings.
Deliverables (In-scope)	Will include a responsive TMVisuals website and flutter mobile app prototype containing a homepage, owner information, business information, services, categorised portfolio, booking and enquiry functionality, availability management, booking-status functionality, secure client content access, social media integration and an

	administration area. Supporting documentation will include the Software Requirements Specification, stakeholder findings, system and user-flow documentation, testing documentation and final project documentation.
Acceptance Criteria	- Website and application displays photo and video files in real time - Clients can make booking requests - Clients can view owner/business information
Exclusions (Out-Of-Scope)	- Online payment - Photo Editing - Video Editing
Constraints	- Limited project timeframe - Limited development resources - Unavailability of project budget
Assumptions	- The stakeholder will provide approved photography and videography content. - The stakeholder has permission to display the portfolio content provided - The stakeholder will review the system and provide feedback - Pricing will remain customised and negotiable with a non refundable deposit - Google drive may continue to be used for high-quality content delivery while in developing the in web/app storage

6. StakeHolder Needs & Engagement Findings

6.1 Key findings confirmed during the meeting:

Stakeholder needs a professional online presence, a centralised portfolio,better booking management, availability tracking, protection against double-booking, flexible/customised pricing , non-refundable deposit policy to be clearly communicated, secure private client-content delivery, a centralised way to manage enquiries and questions. Simple administration tools.

Stakeholders want continuous communication with the development team, technical problems to be reported and resolved collaboratively.

7. Functional Requirements:

ID	Functional Requirement	Stakeholder Need
FR-01	The system shall display information about TM Visuals, including the business, services offered and contact information.	Professional centralised online presence
FR-02	The system shall allow users to view an approved public portfolio of photography and videography work.	Clients can view previous work
FR-03	The system shall organise portfolio content into categories such as Food, Graduation, Weddings/Parties and Branding.	Easier navigation of previous work
FR-04	The system shall provide links to TM Visuals' approved social-media platforms.	Maintain existing digital presence
FR-05	The system shall allow potential clients to submit a booking.	Simplify the current messaging-based booking process
FR-06	The booking shall capture the event date, event type, location, duration and other relevant requirements.	Information currently gathered manually by the photographer
FR-07	The system shall allow clients to view available and unavailable booking dates.	Improve availability management
FR-08	The system shall prevent a confirmed booking date from being booked by another client.	Prevent double-booking
FR-09	The system shall allow the administrator to review incoming booking requests.	Owner must manage bookings
FR-10	The system shall allow the administrator to approve, reject or update a booking request.	Owner retains control over booking decisions
FR-11	The system shall allow clients to view the current status of their booking	Improve client communication

	request.	
FR-12	The system shall display the requirement for a 50% non-refundable deposit before a booking is confirmed.	Stakeholder's booking policy
FR-13	The system shall allow the administrator to upload and manage approved public portfolio content.	Owner needs simple content-management tools
FR-14	The system shall distinguish between public portfolio content and private client content.	Protect client privacy
FR-15	The system shall provide authenticated clients with access to their own completed photographs or videos.	Secure content delivery
FR-16	The system shall allow authorised clients to download their completed content or access the corresponding secure delivery link.	Existing Google Drive/content-delivery process
FR-17	The administrator shall be able to associate completed client content with the relevant client or booking.	Organised content management
FR-18	The administrator shall be able to view and manage client and booking information through an administration interface.	Reduce manual administration
FR-19	The system shall store client contact information supplied during bookings so that TM Visuals can communicate with the client.	Continued client communication

8. Non-Functional Requirements:

ID	Category	Requirement
NFR-01	Usability	The website and mobile application shall provide a simple interface that can be used by clients without technical training.
NFR-02	Usability	Administrative functions shall be designed so that the TM Visuals owner can manage bookings and portfolio content without development knowledge.

NFR-03	Compatibility	The website shall use a responsive layout that functions correctly on commonly used desktop, tablet and mobile screen sizes.
NFR-04	Performance	Normal pages and portfolio information should load within approximately 3 seconds under a normal internet connection, excluding delays caused by large external media files.
NFR-05	Performance	Portfolio images shall be compressed or optimised before delivery to reduce unnecessary bandwidth and application load.
NFR-06	Security	Private client content shall only be accessible to the authenticated client associated with that content and authorised administrators.
NFR-07	Security	The system shall enforce authentication and authorisation before allowing access to administrative functionality or private client information.
NFR-08	Security	Sensitive credentials, API keys and database secrets shall not be stored in publicly accessible source-code repositories or client-side code.
NFR-09	Privacy	Personal information shall only be collected and used for legitimate booking, communication and content-delivery purposes.
NFR-10	Privacy	Public portfolio media and private client media shall be stored and accessed separately so that private client content cannot accidentally be exposed publicly.
NFR-11	Data Integrity	Booking updates shall preserve consistent availability information across the mobile application and website.
NFR-12	Reliability	A confirmed booking shall remain recorded consistently in the shared database and be visible to all authorised system interfaces.
NFR-13	Availability	The website, mobile application and backend services shall be available whenever the required hosting and Supabase services are operational and an internet connection is available.
NFR-14	Maintainability	Application components shall be organised so that changes to the Flutter application, ASP.NET backend and Supabase database can be maintained without unnecessarily affecting unrelated components.

NFR-15	Quality	Error messages shall clearly inform users when an operation such as login, booking submission, media access or data retrieval fails.

9. Integration Requirements

ID	Title	Description
IR-01	Shared Backend	The Flutter mobile application and ASP.NET web application shall operate against the same logical backend and shared Supabase data source so that booking, client, availability and content information remains consistent across both applications.
IR-02	ASP.NET API	The ASP.NET backend shall expose the application operations required by the frontend systems, including booking management, availability management, client information and administrative operations.
IR-03	Flutter Integration	The Flutter mobile application shall communicate with the backend services through secured API requests and shall display data retrieved from the shared system.
IR-04	Supabase Database	Supabase shall store persistent application data such as users, booking requests, booking status, availability information, client information and portfolio metadata.
IR-05	Supabase Storage	Supabase Storage may be used to store approved portfolio media and other system-managed media where appropriate.
IR-06	Authentication and Authorisation	Authentication information shall be used across the integrated system to identify users and determine whether they are permitted to access client or administrative resources.

IR-07	Data Synchronisation	When booking or availability information is changed through an authorised application, the updated information shall be stored centrally so that the website and mobile application retrieve the same current data.
IR-08	Private Media Access	Private media stored within the system shall use controlled access mechanisms so that the Flutter application and web application only expose media to the authorised client and administrator.
IR-09	Google Drive Integration	Where high-quality completed client media continues to be delivered through Google Drive, the system shall allow an authorised Google Drive delivery link to be associated with the relevant client's booking or content record.
IR-10	Social Media Integration	The website and mobile application shall provide links to the approved TM Visuals Instagram and TikTok accounts.
IR-11	WhatsApp Integration	The system may provide a link or contact mechanism allowing users to initiate communication with TM Visuals through WhatsApp while structured booking information remains stored in the central system.
IR-12	Error Handling	If an external integration such as Supabase or Google Drive is unavailable, the system shall report the failure to the user without exposing credentials or confidential technical information.

10. Feasibility Study

Feasibility Area	**To Be Evaluated**
Technical feasibility	The system can be developed using Flutter, ASP.NET, and Supabase.

	The group already has experience with these technologies, which reduces development risks and makes the project more achievable within the available timeframe. Any additional features can be supported through further research and learning if necessary.
Operational feasibility	The proposed system addresses the main challenges faced by TM Visuals, including the lack of a booking system, poor communication management, and limited visibility of the company's portfolio. Since the stakeholder is familiar with digital tools, the adoption of the system is expected to be smooth. The system should also support flexible pricing options instead of fixed rates.
Economic feasibility	The project is cost-effective because Flutter, ASP.NET, and Supabase offer free tiers that are sufficient for developing the prototype. As a result, there are no significant software or infrastructure costs involved. The primary investment required is the time and effort of the group members.
Schedule feasibility	The project is realistic within the available semester timeframe. Effective task allocation among the eight group members will help ensure completion on schedule. However, any new requirements introduced after the stakeholder meeting may affect the project timeline.
Legal feasibility	The system will store personal information and client-related content, which means it must comply with POPIA regulations. Appropriate access controls should be implemented to protect private content, and personal information should only be collected and used for its intended purpose.
Ethical feasibility	The stakeholder expressed concerns about losing control over shared images and content. Therefore, the system should clearly distinguish between publicly available portfolio content and private client content. The development team must also ensure that all system features are based on confirmed stakeholder requirements and avoid making unnecessary assumptions.

Feasibility Conclusion:

Overall, the proposed system is feasible and can be completed within the available time and resources. The group has experience with the technologies needed to develop the system, and the system will help solve the main problems faced by TM Visuals. The project will also have low costs because free versions of the required tools can be used. Privacy and POPIA requirements will need to be considered when handling client information and photos. Any changes to the requirements may affect the project schedule, so they will need to be discussed with the stakeholder.

11. Project Risk Register

Risk ID	**Risk Category**	**Risk Description**	**Likelihood**	**Impact (1-5)**	**Risk Score Level(1-5)**	**Mitigation action**	**Contingency action**	**Owner**	**Status**
RSK-001	Technical/Performance	Tries uploading high quality footage which crashes the app due to file size limits	5	4	20	Compress the files before sending them to Supabase storage	Set strict upload file sizes	Lead Dev	Open
RSK-002	Project Management	The project scope may expand if the group includes additional features that were not agreed upon by the stakeholder.	3	4	12	Stick strictly to confirmed requirements; log new ideas in change log.	Review scope weekly.	Person 4	Open

RSK-003	Requirement	Not or short understanding the stakeholder requirement.	2	4	8	Confirm requirements in the notes after the meeting.	Following up with the stakeholder before you start developing	Person 7	Open
RSK--004	Team	Group members unavailable during critical work periods	3	3	9	Clear deadlines and do regular check-ins	Redistribute tasks among available members	Project manager	Open
RSK-005	Documentation	Inconsistent formatting/ content across sections when merging SRS	3	2	6	Choose one team member to check the whole document at the end and make sure the formatting and content are consistent.	Redo formatting before submission if issues found late	Person 5	Open
RSK-006	Project Management	GitHub repository poorly maintained, missing commit evidence	2	3	6	Set repo structure early; require regular commits	PM reviews repo weekly, reminds members	Project Manager	Open

RSK-007	Project Management	Scope creep as group adds features beyond what stakeholder confirmed	3	4	12	Stick strictly to confirmed requirements; log new ideas in change log	Review scope weekly; cut non-essential features if behind	Person 4	Open
RSK-008	Technical	Integration issues between Flutter, ASP.NET, and Supabase	3	4	12	Build small integration prototypes early	If the system doesn't work properly, make the integration easier or turn off some real-time features to keep it stable.	Person 2	Open
RSK-009	Security	Private client content accidentally exposed publicly	2	5	10	Design and test access-control rules carefully	Immediately restrict content and notify stakeholder	Person 3	Open
RSK-010	Schedule	MS Project schedule becomes unrealistic as work progresses	3	3	9	Review and update schedule regularly	Adjust milestones and reprioritize tasks	Person 6	Open

RSK-011	Technical	Application may experience bugs or crashes during testing due to unexpected errors	3	4	12	Conduct regular testing throughout development and fix errors as they are identified.	Prioritize critical bugs and disable affected features temporarily if necessary.	Person 1	Open
RSK-012	Team	Poor communication between group members may result in duplicated work or important tasks being missed	3	3	9	Use regular meetings and clearly assign tasks and responsibilities to each member	Reassign missed or duplicated tasks and clarify responsibilities with the team	Project Manager	Open

Risk Score = Impact * Likelihood

Top 3 highest project risks:

#1 RSK-001

#2 RSK-002

#3 RISK-007

12. Requirements Change Log

Change ID	Date	Requirements / Section Affected	Before	After	Reason for change	Requested by	Approved by	Status
CL-001	2026-08-30	Pricing model	Early scope	Administrator enters	Stakeholder clarified price	Stakeholder (T	Project Team	implemente

		(FR-12)	implied a fixed/rigid price per event type.	a customised quotation per client, no fixed pricing.	depends on location,scope and client budget, not event category (Interview S-7, S-22)	Mulaudzi)		d
CL-002	2026-08-30	Content delivery (IR-09)	Delivery method not defined in early scope.	Google Drive integration retained for high-quality delivery, linked to client's booking	Stakeholder confirmed google drive will continue to be used while in-app storage is developed (Assumptions S-13,Interview S-17)	Stakeholder	Project Team	implemented
CL-003	2026-08-31	Public vs private content (FR-16, FR-17, NFR-06, NFR-10)	No distinction between public and private content.	Added requirement to separate and protect private client content from the public portfolio.	Stakeholder raised concerns about losing control of images once shared (Interview S-13, S-19, S-20)	Stakeholder	Project Team	implemented
CL-004	2026-08-31	Availability and double booking (FR-07, FR-08,NRF-11,NRF-12)	Not addressed in early scope.	Added availability calendar and logic to prevent a confirmed date being booked twice.	Stakeholder identified double booking as a major current risk (Problem Statement S-3, interview S-21)	Stakeholder	Project Team	implemented
CL-005	2026-09-01	Deposit and terms disclosure (FR-13,FR-14)	Not specified.	Added requirement to display the 50% non-refundable deposit policy and	Stakeholder stressed the policy must be clearly communicated to avoid disputes (Interview S-8, S-9)	Stakeholder	Project Team	implemented

				Terms & Conditions before booking is finalised.				
CL-006	2026-09-01	Portfolio categorisation (FR-03)	General portfolio only, no structure.	Portfolio organised into Food, Graduation, Weddings/ Parties and Branding categories.	Stakeholder specified these categories for easier navigation (Interview S-15)	Stakeholder	Project Team	implemented
CL-007	2026-09-03	Project title and framing (S-5.2 vs S-13)	“TM Visual”	"TM Visuals Photography, Videography and Client Management System"	Refined to reflect the full booking and content management system, not just the brand name.	Project Team	Pending stakeholder sign-off	implemented
CL-008	2026-09-03	Exclusions (S-5.2 vs S-13)	Limited to Online Payment, Photo Editing, Video Editing	Expanded to also exclude automated pricing decisions, fixed pricing, automated content creation, and any user roles beyond client/administrator.	Prevent scope creep flagged in Risk Register (RSK-002,007); clarify boundaries after further discussion.	Project Team	Project Team	implemented
CL-009	2026-09-04	Constraints (S-5.2 vs S-13)	General constraints only (timeframe, resources, budget).	Added constraints on internet dependency, content quality depending on stakeholde	Derived from Feasibility Study and Interview findings on the owner’s technical ability.	Project Team	Project Team	implemented

				r supply, need for a non-technical-friendly system, and third-party service limitations.				
CL-010	2026-09-04	Assumptions (S-5.2 vs S-13)	Several granular assumptions, incl. continued Google Drive use and stakeholder review/feedback.	Consolidated into two higher-level assumptions: business continuity and accuracy of stakeholder-provided content/information.	Simply for clarity; Google Drive dependency moved under Constraints instead.	Project team	Project Team	implemented
CL-011	2026-09-05	FR-12	The system shall allow the administrator to enter a customised quotation for a client rather than automatically assigning a fixed price.	The system will allow the administrator to enter a customized invoice for a client.				Proposed
CL-012								Proposed

13. Refined Project Scope

Project title	TM Visuals Photography, Videography and Client Management System
Project Aim	The aim of the project is to develop a professional website and mobile application for TM Visuals that provides a central place for clients to view the portfolio, make enquiries, manage bookings, check availability and securely access their completed photographs and videos. It will allow the administrator to view and manage client information.
Objectives	Create a professional online presence for TM Visuals. Allow potential clients to view previous photography and videography work. Organise the portfolio into categories such as Food, Graduation, Weddings/Parties and Branding. Allow clients to view frequently asked questions. Collect important event information such as date, location, event type and duration. Help the owner manage bookings and availability. Reduce the risk of double-booking. Communicate the 50% deposit requirement. Allow the owner to provide customised prices and quotations. Provide secure access to private client content. Improve the quality and organisation of client content delivery.
Refined Scope Overview	The project will focus on developing a centralised TM Visuals website. The website will include business information, services, portfolio, booking, availability and client content features. Clients will be able to view public portfolio content without needing social media. Existing clients will be able to access and download their private completed work.

	The owner will be able to manage bookings, portfolio content and client information. Pricing will remain customised and will not be displayed as fixed prices.
Deliverables	- A functional TM Visuals website. - Homepage and business information. - Services section. - Categorised portfolio. - Booking-status feature. - Secure client content access. - Social media links. - Flutter mobile application. - ASP.NET backend/API. - Supabase database and storage. - Client booking system. - Availability calendar. - Admin dashboard. - Secure client-content area. - Portfolio management. - Terms and Conditions page. - User/system flow diagrams. - Testing and final project documentation.
Acceptance Criteria	- Clients can access website and application - Clients can view business information and services offered - Website and application uploads and renders compressed portfolio photos via superbase Storage - Portfolio content is organised into categories - Clients can submit booking - Clients can negotiate prices - The 50% non-refundable deposit is communicated - Owner can view and manage booking requests and enquiries - High-quality clients can access their content - Portfolio content can be stored, retrieved and accessed
Exclusions	The system will not edit photographs or videos. The system will not replace Photoshop or CapCut. The system will not automatically decide prices for the owner. The system will not use fixed prices for all services. The system will not automatically create photography or videography content. The system will not make business decisions on behalf of

	the owner. Private client content will not be available to the general public No unnecessary user roles beyond client and administrator.
Constraints	The project must be completed within the available project timeframe. The system requires an internet connection. The quality of the portfolio depends on content provided by TM Visuals. The owner is not a technical developer, so the system must be simple to manage. Pricing must remain flexible because every client may have different requirements. Private client information and media must be protected. External services such as Google Drive and payment systems may have technical limitations. The project must remain within the available resources.
Assumptions	The project assumes that TMVisuals will continue operating under the same business name and that the owner will provide accurate business information, portfolio content and client information

PHASE 2: SYSTEM ARCHITECTURE AND DESIGN

14. Links to Project work and Diagrams

14.1 WorkSpace and Database Links

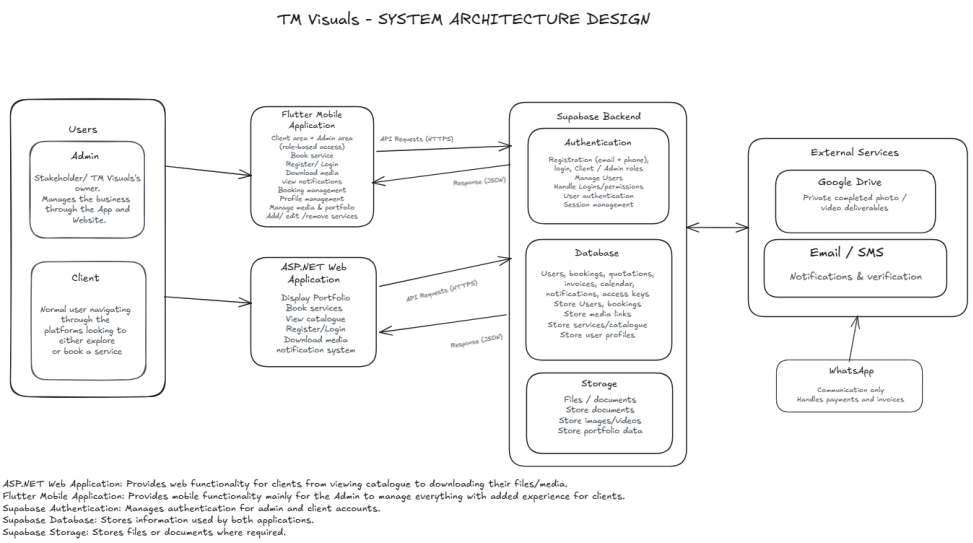
Github Project repo:
https://github.com/Truce-hash/TM-Visuals-Project

Supabase Database:
https://supabase.com/dashboard/project/fswyeuiajsbdlkhhydnx

14.2 System Diagram Links

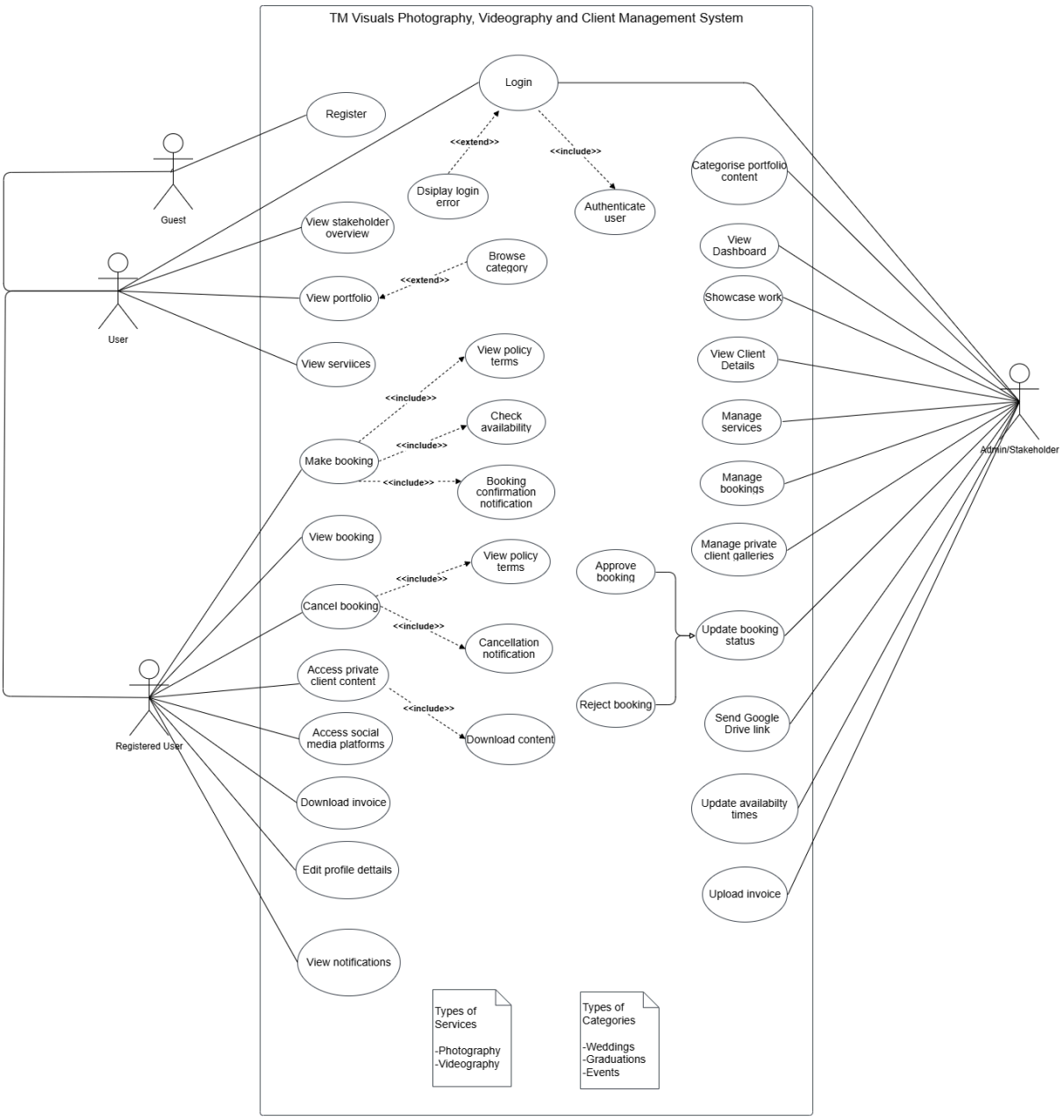
Link	Tool Used	Responsibility	File
System Architecture Diagram	Excalidraw	222048665, 224107046	
TM Visuals Use Case	draw.io	224051673, 222045625	Use Case Design File
ASPNET Website Sequence Diagram	draw.io	224090026	
Flutter Mobile App Sequence Diagram	draw.io	224022456	
ERD Diagram	draw.io	223021599, 224111760	TM Visuals ERD File
ASP.NET Interface Design	Figma, Penpot	224022456, 224051673, 222045625, 224090026	
Flutter Interface Design	Figma, Penpot	222048665, 224107046, 223021599, 224111760, 224090026	

14.2.1 System Architecture:



ASP.NET Web Application: Provides web functionality for clients from viewing the portfolio to downloading their files/media.
Flutter Mobile Application: Provides mobile functionality mainly for the Admin to manage everything with added experience for clients.
Supabase Authentication: Manages authentication for admin and client accounts.
Supabase Database: Stores information used by both applications.
Supabase Storage: Stores files or documents where required.

14.2.2 Use Case Diagram:



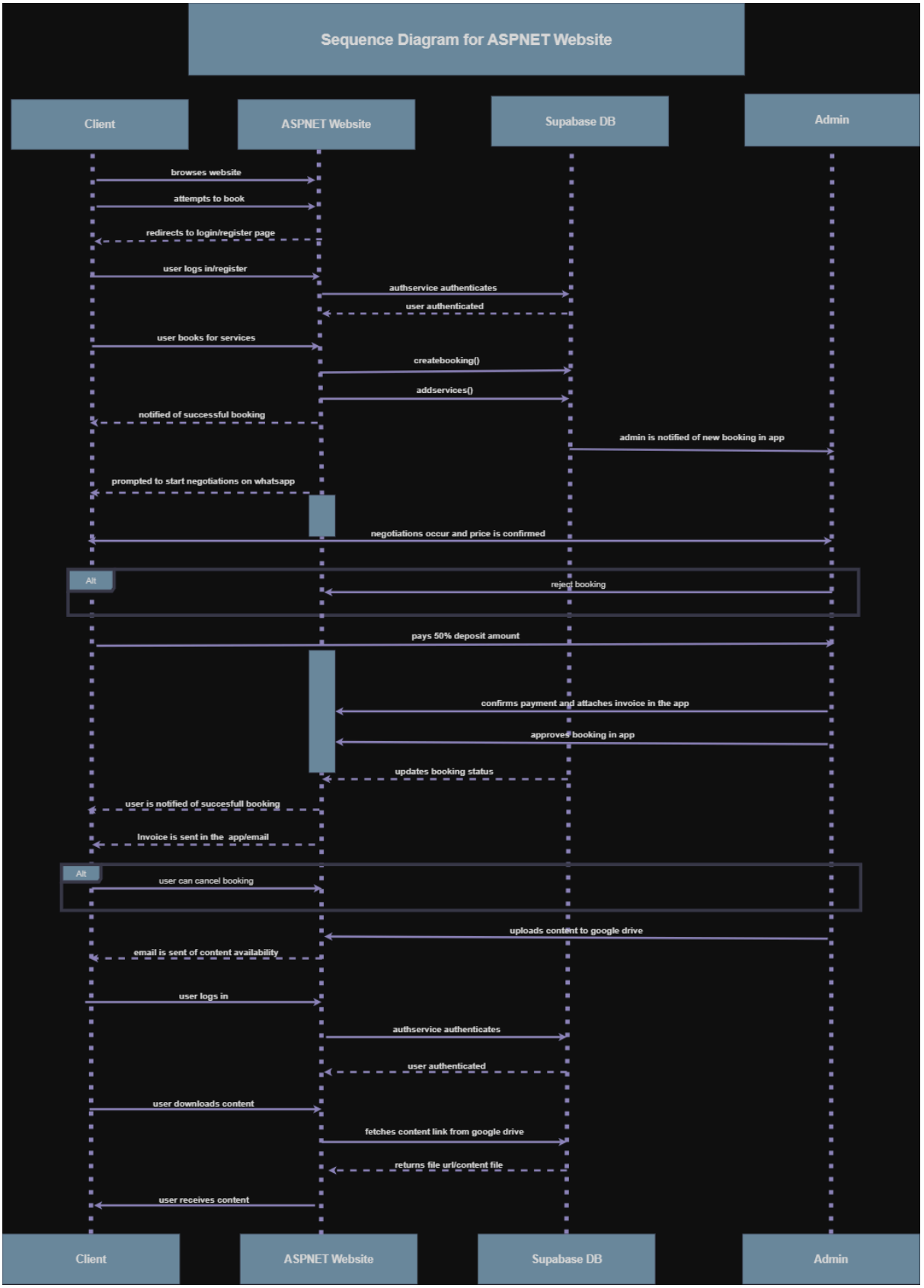
The Use Case Diagram shows the main interactions between the users of the TM Visuals system and the system functions. The main actors are Guest, Registered User, Client and Administrator/Stakeholder.

Guests can register and log in to the system. General Users can view the stakeholder overview, view the portfolio, browse categories, and view services. Registered Users can make bookings, which involves viewing policy terms, checking availability, and receiving booking confirmation notifications. They can also view bookings and cancel bookings, which includes viewing policy terms and receiving cancellation notifications. Furthermore, they are able to access private client content

to download media, access social media platforms, download invoices, edit profile details, and view notifications. Administrators/Stakeholders log in through a process that includes user authentication and login error displays. They manage the system dashboard, categorise portfolio content, showcase work, view client details, and manage services, bookings, and private client galleries. Finally, they handle backend workflows by updating booking statuses through approvals or rejections, sending Google Drive links, updating availability times, and uploading invoices.

14.2.3 Sequence Diagrams:

14.2.3 ASP.NET Web App Diagram:



ASP.NET Sequence Diagram Summary

The diagram illustrates the sequence of interactions between the participants, showing who communicates with whom and in what order.

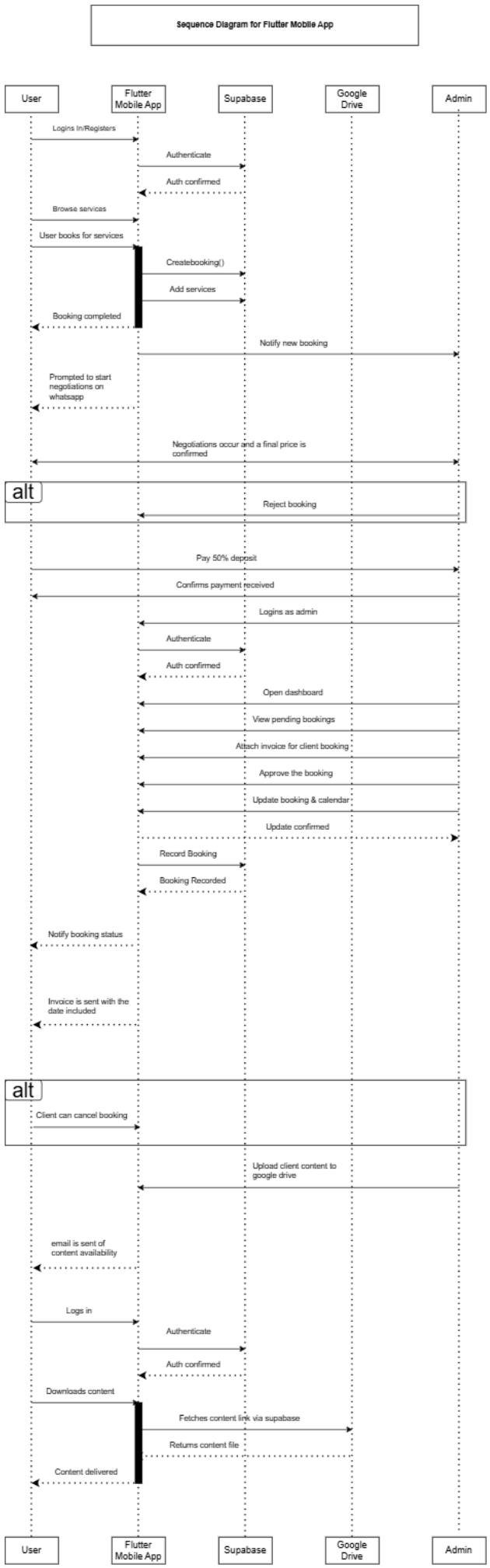
The user explores the website and browses the available services, portfolio, and other content. When attempting to make a booking, the user is redirected to the registration or login page to create an account or sign in before proceeding. Once logged in, the user checks the calendar for available time slots and proceeds with the booking. Before submitting the booking request, the user must agree to the terms and conditions, which state that a non-refundable 50% deposit is required to secure the booking.

The user is then prompted to initiate negotiations via WhatsApp, where the price and payment arrangements are discussed. Meanwhile, the administrator receives the booking request through the mobile application, along with a WhatsApp message containing the user's details and booking information.

Once payment has been confirmed, the administrator sends the user an invoice, approves the booking, and updates its status. The user is then notified that the booking has been successfully confirmed and receives the invoice via both email and the mobile application.

Finally, once the content has been uploaded and made available for retrieval, the user receives an email notification. The user can then log in to the system and download the content.

14.2.4 Flutter Mobile App Diagram:


Sequence Diagram for Flutter Mobile App
User
Flutter Mobile App
Supabase
Google Drive
Admin
Logins In/Registers
Authenticate
Auth confirmed
Browse services
User books for services
Createbooking()
Add services
Booking completed
Notify new booking
Prompted to start negotiations on whatsapp
Negotiations occur and a final price is confirmed
alt
Reject booking
Pay 50% deposit
Confirms payment received
Logins as admin
Authenticate
Auth confirmed
Open dashboard
View pending bookings
Attach invoice for client booking
Approve the booking
Update booking & calendar
Update confirmed
Record Booking
Booking Recorded
Notify booking status
Invoice is sent with the date included
alt
Client can cancel booking
Upload client content to google drive
email is sent of content availability
Logs in
Authenticate
Auth confirmed
Downloads content
Fetches content link via supabase
Returns content file
Content delivered
User
Flutter Mobile App
Supabase
Google Drive
Admin

Flutter Mobile App Sequence Diagram: Explanation

This diagram shows the full booking process in the mobile app, from a client browsing services to receiving their finished photos. It has five participants: the User (client), the Flutter Mobile App, Supabase, Google Drive and the Admin (the stakeholder).

The client browses services, then logs in or registers. The app checks the details with Supabase. The client submits a booking, which is saved in Supabase, and the admin is notified. The admin logs in, opens the dashboard, views the pending bookings and approves one. The price is then negotiated with the client on WhatsApp. An alt frame shows the booking can be rejected instead. Once a price is agreed, the client pays the 50% deposit and the admin confirms it. The invoice is attached, the booking and calendar are updated, and the client sees the confirmation, the date and the invoice. A second alt frame shows the client can still cancel. When the shoot is finished, the admin uploads the content to Google Drive, the client is emailed, logs in and downloads it.

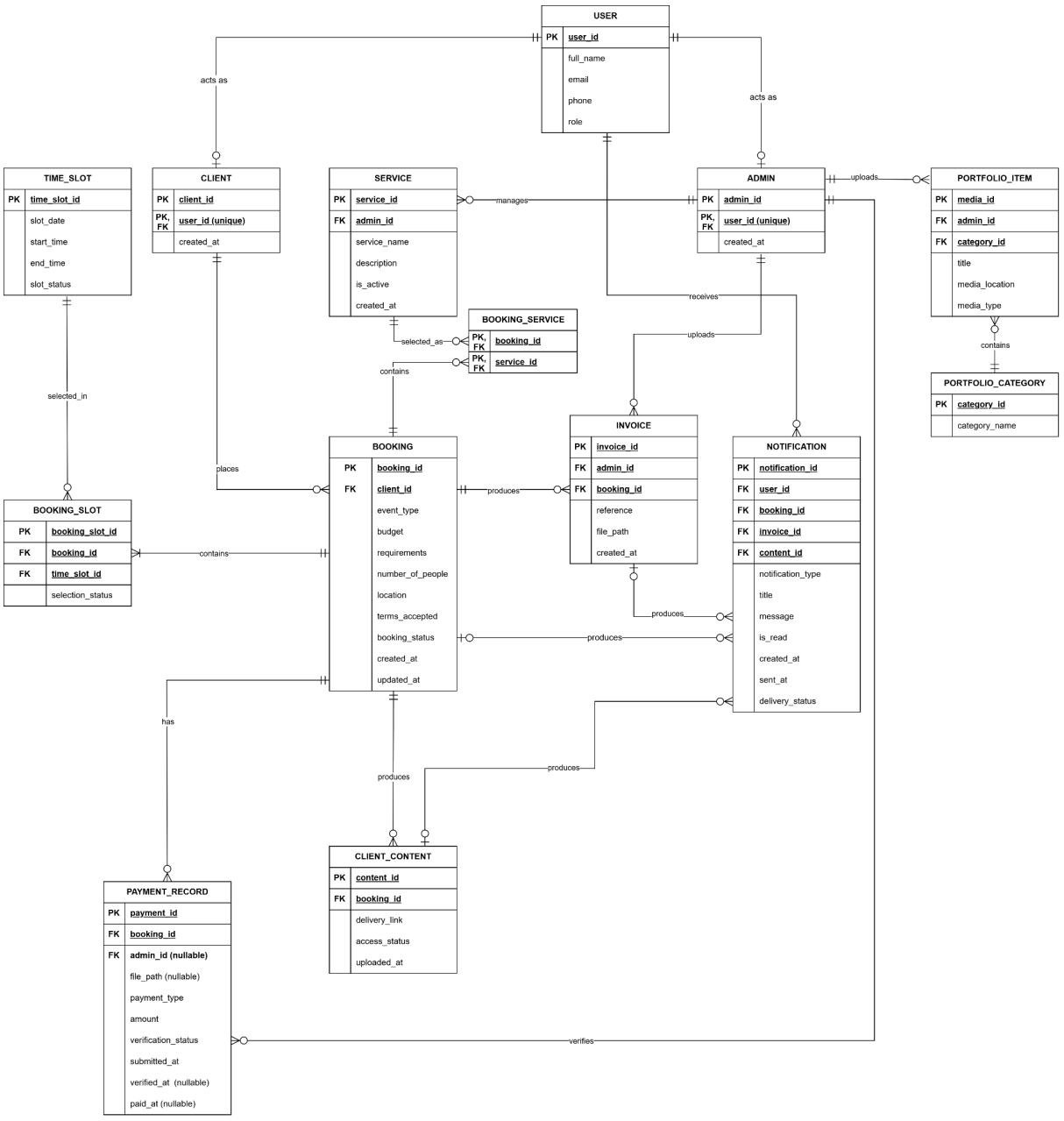
15. DATABASE DESIGN: ENTITY RELATIONSHIP DIAGRAM (ERD)

ERD Diagram:

15.1 Introduction

The Entity Relationship Diagram (ERD) shows the proposed TM Visuals database structure for users, client and administrator records, services, bookings, scheduling, payment records, invoice files, private client content, public portfolio items and notifications.

The database is designed to support the information captured and displayed through the Flutter application and ASP.NET interfaces. Primary keys uniquely identify records, while foreign keys establish relationships between related entities. The design aims to maintain data consistency, support efficient retrieval of information and facilitate the management of bookings and client services.



15.2 Description of Entities and Attributes

15.2.1 USER

The USER entity stores the basic information of individuals registered on the system. It contains the attributes user_id, full_name, email, phone and role. The user_id is the primary key and uniquely identifies each user. The role attribute distinguishes between clients and administrators, allowing the system to apply different access permissions.

15.2.2 CLIENT

The CLIENT entity identifies clients who place bookings. Its attributes are client_id, user_id and created_at. The diagram marks client_id as a primary-key attribute and user_id as both a primary-key and foreign-key attribute referencing USER.user_id. The created_at attribute

records when the client record was created. Each client record belongs to one user, and a user may have zero or one client record.

15.2.3 ADMIN

The ADMIN entity identifies administrators responsible for managing services, uploading portfolio content, attaching invoices and verifying payments. Its attributes are admin_id, user_id and created_at. The diagram marks admin_id as a primary-key attribute and user_id as both a primary-key and foreign-key attribute referencing USER.user_id. The created_at attribute records when the administrator record was created. Each administrator record belongs to one user, and a user may have zero or one administrator record.

15.2.4 SERVICE

The SERVICE entity stores the services offered by TM Visuals. Its attributes are service_id, admin_id, service_name, description and is_active. The service_id is the primary key, while admin_id references ADMIN.admin_id and identifies the responsible administrator. The is_active attribute indicates whether a service is available for selection. Services are associated with bookings through BOOKING_SERVICE.

15.2.5 TIME_SLOT

The TIME_SLOT entity stores calendar periods associated with booking-slot selections. Its attributes are time_slot_id, booking_slot_id, slot_date, start_time, end_time and slot_status. The time_slot_id is the primary key, while booking_slot_id references BOOKING_SLOT.booking_slot_id. The slot_date, start_time and end_time identify the calendar period, and slot_status records its availability state. In this ERD, one BOOKING_SLOT record can be associated with zero or many TIME_SLOT records, and each TIME_SLOT references one BOOKING_SLOT.

15.2.6 BOOKING

The BOOKING entity records booking requests and their details. Its attributes are booking_id, client_id, event_type, budget, requirements, number_of_people, location, terms_accepted, booking_status, created_at and updated_at. The booking_id is the primary key, while client_id references CLIENT.client_id.
The event_type, number_of_people and location describe the event. The budget records the client's proposed budget, and requirements stores additional request details. The terms_accepted attribute records acceptance of the booking terms. The booking_status records the current workflow state, while created_at and updated_at record creation and the latest update. BOOKING connects clients to selected services, scheduling records, payments, invoice files, notifications and completed content.

15.2.7 BOOKING_SERVICE

The BOOKING_SERVICE entity acts as an associative entity between BOOKING and SERVICE, implementing a many-to-many relationship. It contains two attributes: booking_id and service_id. Both attributes are foreign keys referencing the BOOKING and SERVICE entities, respectively, and together form a composite primary key.

The composite primary key ensures that a particular service can only be associated with a booking once. This structure allows a booking to include multiple services and allows each service to be associated with multiple bookings without creating duplicate associations.

15.2.8 BOOKING_SLOT

The BOOKING_SLOT entity records scheduling selections associated with bookings. Its attributes are booking_slot_id, booking_id and selection_status. The booking_slot_id is the primary key, while booking_id references BOOKING.booking_id. The selection_status records the state of the scheduling selection.
Each BOOKING_SLOT belongs to one booking. TIME_SLOT records reference BOOKING_SLOT through booking_slot_id, allowing calendar periods to be associated with the selection. Availability checks and coordinated updates are required to prevent conflicting confirmed reservations.

15.2.9 PAYMENT_RECORD

The PAYMENT_RECORD entity stores recorded payments associated with bookings. Its attributes are payment_id, booking_id, admin_id, payment_type, amount, verification_status, submitted_at, verified_at and paid_at. The payment_id is the primary key, while booking_id references BOOKING.booking_id.
The nullable admin_id references ADMIN.admin_id and identifies the verifying administrator when assigned. The payment_type identifies the kind of payment, such as a deposit or balance payment, and amount records its value. The verification_status records the current verification state. The submitted_at attribute records submission time; verified_at and paid_at are nullable and record verification time and the payment date when known. This entity records payments made outside the system. It does not contain a payment-proof file reference in the supplied ERD.

15.2.10 INVOICE

The INVOICE entity stores references to invoice files associated with bookings. Its attributes are invoice_id, admin_id, booking_id and file_path. The invoice_id is the primary key. The admin_id references ADMIN.admin_id and identifies the administrator who attached the invoice, while booking_id references BOOKING.booking_id.
The file_path identifies the stored invoice file. The database stores the association and file reference rather than separate invoice line items, amounts, balances or issue dates. A booking can have zero or many invoice records.

15.2.11 CLIENT_CONTENT

The CLIENT_CONTENT entity stores references to completed photography or videography content delivered to clients. Its attributes include content_id, booking_id, delivery_link, access_status and uploaded_at.
The content_id is the primary key, while booking_id references BOOKING. The delivery link provides a reference to the location of the delivered content, and the access status indicates whether the content is available or restricted. This entity supports the delivery of completed work while helping protect the privacy of client photographs and videos.

15.2.12 PORTFOLIO_CATEGORY

The PORTFOLIO_CATEGORY entity groups portfolio items into categories. It contains category_id as the primary key and category_name to identify each category. Categories help organise photography and videography work so that visitors can browse relevant examples of the business's services.

15.2.13 PORTFOLIO_ITEM

The PORTFOLIO_ITEM entity stores information about photographs and videos selected for the public portfolio. Its attributes include media_id, admin_id, category_id, title, media_location and media_type.
The media_id is the primary key. The admin_id references ADMIN and identifies the administrator responsible for uploading the media, while category_id references PORTFOLIO_CATEGORY. The media location identifies where the file is stored, and the media type distinguishes between images and videos. This entity supports the presentation and management of approved portfolio content.

15.2.14 NOTIFICATION

The NOTIFICATION entity stores notifications intended for registered users. Its attributes are notification_id, user_id, booking_id, invoice_id, content_id, notification_type, title, message, is_read, created_at, sent_at and delivery_status.
The notification_id is the primary key. The user_id is a foreign key referencing USER and identifies the intended recipient. The booking_id, invoice_id and content_id are foreign keys referencing BOOKING, INVOICE and CLIENT_CONTENT, respectively, and associate a notification with the relevant record. Not every notification needs every business reference; the applicable reference depends on the notification type.
The notification_type identifies the event, while title and message provide the notification details. The is_read attribute records whether the recipient has read the notification. The created_at, sent_at and delivery_status attributes support tracking of notification creation and delivery. This entity supports booking updates, invoice alerts and notifications that completed content is available.

16. Relationships Between Entities

1. USER-CLIENT: A user may act as zero or one client, and each client record belongs to one user.
2. USER-ADMIN: A user may act as zero or one administrator, and each administrator record belongs to one user.
3. CLIENT-BOOKING: A client can place zero or many bookings, while each booking belongs to one client.
4. ADMIN-SERVICE: An administrator can manage zero or many services, while each service references one administrator.
5. BOOKING-SERVICE: These entities have a many-to-many relationship implemented through BOOKING_SERVICE. Each association references one booking and one service. Its composite primary key prevents duplicate associations.
6. BOOKING-BOOKING_SLOT: A booking contains one or more booking-slot records, and each booking-slot record belongs to one booking.
7. BOOKING_SLOT-TIME_SLOT: A booking-slot record can be associated with zero or many time-slot records, while each time slot references one booking-slot record.

8. BOOKING-PAYMENT_RECORD: A booking can have zero or many payment records, while each payment record belongs to one booking.
9. ADMIN-PAYMENT_RECORD: An administrator can verify zero or many payment records. The nullable admin_id allows a payment record to have no verifying administrator until one is assigned.
10. BOOKING-INVOICE: A booking can have zero or many invoice records, while each invoice references one booking.
11. ADMIN-INVOICE: An administrator can attach zero or many invoice files, while each invoice record references one administrator.
12. BOOKING-CLIENT_CONTENT: A booking can produce zero or many client-content records, while each content record belongs to one booking.
13. ADMIN-PORTFOLIO_ITEM: An administrator can upload zero or many portfolio items, while each portfolio item references one administrator.
14. PORTFOLIO_CATEGORY-PORTFOLIO_ITEM: A category can contain zero or many portfolio items, while each portfolio item belongs to one category.
15. USER-NOTIFICATION: A user can receive zero or many notifications, while each notification has one recipient.
16. BOOKING-NOTIFICATION: A booking can be associated with zero or many notifications. The diagram allows a notification to have no booking reference or to reference one booking.
17. INVOICE-NOTIFICATION: An invoice can be associated with zero or many notifications. The diagram allows a notification to have no invoice reference or to reference one invoice.
18. CLIENT_CONTENT-NOTIFICATION: A client-content record can be associated with zero or many notifications, while each notification on this relationship references one client-content record, as drawn in the ERD.

17. Primary Keys and Foreign Keys

Primary keys uniquely identify records. The principal identifiers shown are user_id, client_id, admin_id, service_id, time_slot_id, booking_slot_id, booking_id, payment_id, invoice_id, content_id, category_id, media_id and notification_id. BOOKING_SERVICE uses booking_id and service_id together as a composite primary key. CLIENT and ADMIN also mark user_id as a primary-key attribute as well as a foreign key in the supplied diagram.
Foreign keys connect related records. CLIENT.user_id and ADMIN.user_id reference USER.user_id. SERVICE.admin_id, INVOICE.admin_id and the nullable PAYMENT_RECORD.admin_id reference ADMIN.admin_id. BOOKING.client_id references CLIENT.client_id.
BOOKING_SERVICE references BOOKING and SERVICE. BOOKING_SLOT.booking_id references BOOKING.booking_id, while TIME_SLOT.booking_slot_id references BOOKING_SLOT.booking_slot_id. PAYMENT_RECORD, INVOICE and CLIENT_CONTENT each reference their associated booking.
PORTFOLIO_ITEM references its administrator and portfolio category.
NOTIFICATION.user_id identifies the recipient, while booking_id, invoice_id and content_id reference the relevant business records. Foreign-key constraints must use referenced identifiers that are primary keys or otherwise unique.

18. Database Design and System Requirements Alignment

The database supports the shared data required by the Flutter mobile application and ASP.NET web application. USER, CLIENT and ADMIN store account and role-related information. SERVICE, PORTFOLIO_CATEGORY and PORTFOLIO_ITEM support service selection and the display of approved public portfolio content.
BOOKING stores client requests, budgets, event details, additional requirements and acceptance of booking terms. BOOKING_SERVICE supports selecting multiple services. BOOKING_SLOT and TIME_SLOT support scheduling selections and calendar periods. Booking status and timestamps support tracking of requests and updates.
PAYMENT_RECORD supports recording and administrator verification of externally made payments. INVOICE associates stored invoice files with bookings and the administrator who attached them. This design stores invoice documents rather than structured invoice line items and financial totals.
CLIENT_CONTENT links completed work to bookings and records delivery links and access status. NOTIFICATION supports booking, invoice and content-related messages, including read status and delivery tracking.
Access-control policies must restrict clients to their own booking-related records, notifications and private content. Administrators may manage business records according to their permissions. Approved public portfolio items must remain separate from private client deliveries.
Primary keys, foreign keys, validation rules and coordinated scheduling updates must support data integrity and prevent conflicting confirmed reservations. The ERD does not include separate fields for agreed booking totals, payment-proof files, invoice line items or rescheduling requests; features requiring those values need an explicitly defined implementation approach.

19. User Interface Design and Design Alignment

19.1 Interface Design Approach

The TM Visuals Photography, Videography and Client Management System provides interfaces for clients and administrators. The Flutter mobile application supports clients who want to browse services, make bookings, receive notifications and access completed media. The ASP.NET web application provides public website access and administrative functions for managing services, bookings, clients, portfolio content, invoices and completed client content.

The interface designs were created in Figma in the file titled Group Q – TM Visuals Interface Designs. The Figma file contains two pages:

- Flutter Mobile Interface Design
- ASP.NET Website Interface Design

FIGMA LINK : [Figma Interface Design]

19.2 Main Flutter Mobile Screens

Screen	Purpose	Main information and actions
Welcome/Home Screen	Introduces the TM Visuals application and gives users access to main features.	Navigate to services, portfolio, bookings, profile and login.
Sign Up Screen	Allows a new client to create an account.	Full name, email, phone number, password and registration action.
Login Screen	Allows registered users and administrator to access the system.	Email, password, login action and error message display.
Services Screen	Displays photography and videography services.	Service name, category, description, price and Book Now action.
Portfolio Screen	Displays completed work by category.	Portfolio images/videos, category filters such as weddings, graduations and events.
Availability Screen	Allows a client to check available appointment dates and time slots.	Date selection, available slots and booking continuation action.
Booking Screen	Allows a client to submit a service booking request.	Selected service, event type, date, time slot, number of people, location and acceptance of terms.

My Bookings Screen	Allows clients to view their submitted bookings.	Booking details, booking status, cancellation/rescheduling options and invoice information.
Notifications Screen	Displays booking and content notifications.	Booking confirmation, approval/rejection, payment and content-availability notifications.
Private Content / My Gallery Screen	Gives clients access to completed photography or videography work.	Content title, media preview, download link and access status.
Profile and Edit Details Screens	Allows a client to manage personal details.	Name, email, phone number, password and update action.

19.3 Main ASP.NET Web Pages

Web page	Purpose	Main information and actions
Home Page	Public entry page for TM Visuals.	Services, portfolio highlights, navigation, login and registration links.
Login and Registration Pages	Provide account access for clients and administrators.	Registration and authentication fields.
Services and Portfolio Pages	Allow website visitors to browse services and completed work.	Service details, categories, portfolio media and booking links.
Booking and Availability Pages	Allow clients to make bookings through the website.	Service selection, date, available time slots, event details and booking submission.

Client Profile and My Gallery Pages	Allow clients to manage their account and access deliverables.	Profile details, booking history, private content and download links.
Manage Bookings Page	Allows the administrator to review and process bookings.	Client details, requested services, date/time slot, booking status, approve/reject actions.
Manage Services Page	Allows the administrator to maintain service offerings.	Add, edit, activate or remove service information.
Manage Portfolio Page	Allows the administrator to manage portfolio categories and media.	Media title, category, media type, upload location, add/edit/delete actions.
Manage Clients Page	Allows the administrator to view client information.	Client profile, contact information and booking history.
Invoice, Deposit and Content Upload Pages	Support booking payments and delivery of completed work.	Invoice amount, payment status, proof of payment, upload completed content and Google Drive link.

19.4 Navigation Design

The ASP.NET web application is mainly a client-facing website. Visitors may access the public Home Page, Services Page and Portfolio Page without logging in. However, a visitor must register and log in before they can make a booking, view booking information, receive booking notifications or access private completed content.

The ASP.NET client navigation flow is:

Home Page → View Services / View Portfolio → Select Service → Register or Login → Availability Page → Booking Form → Booking Confirmation → My Bookings → Notifications → My Gallery / Download Content.

The Flutter mobile application requires user authentication before a user can access the application's main features. When the application opens, the user must log in or register. After successful authentication, the user can access the home screen, services, portfolio, bookings, notifications, profile and private content.

The Flutter client navigation flow is:

Launch Application → Login or Register → Home Screen → Services / Portfolio → Availability → Booking Screen → Booking Confirmation → My Bookings → Notifications → Private Content / My Gallery → Download Content.

The administrator navigation flow is:

Administrator Login → Admin Dashboard → Manage Bookings / Manage Calendar / Manage Profile /More.

19.5 Data Supported by the ERD

The information captured and displayed by the Flutter and ASP.NET interfaces is supported by the entities in the ERD. Registration, login and profile screens use the USER, CLIENT and ADMIN entities. Service and portfolio screens use the SERVICE, PORTFOLIO_ITEM and PORTFOLIO_CATEGORY entities.

The booking and availability interfaces use the BOOKING, BOOKING_SERVICE, TIME_SLOT and BOOKING_SLOT entities. Deposit and invoice information is supported by the DEPOSIT_RECORD and INVOICE entities. The private gallery and download interfaces use the CLIENT_CONTENT entity, while booking, invoice and completed-content updates are displayed through the NOTIFICATION entity.

This ensures that all information entered by users, viewed by clients or managed by administrators has a corresponding entity in the Supabase database.

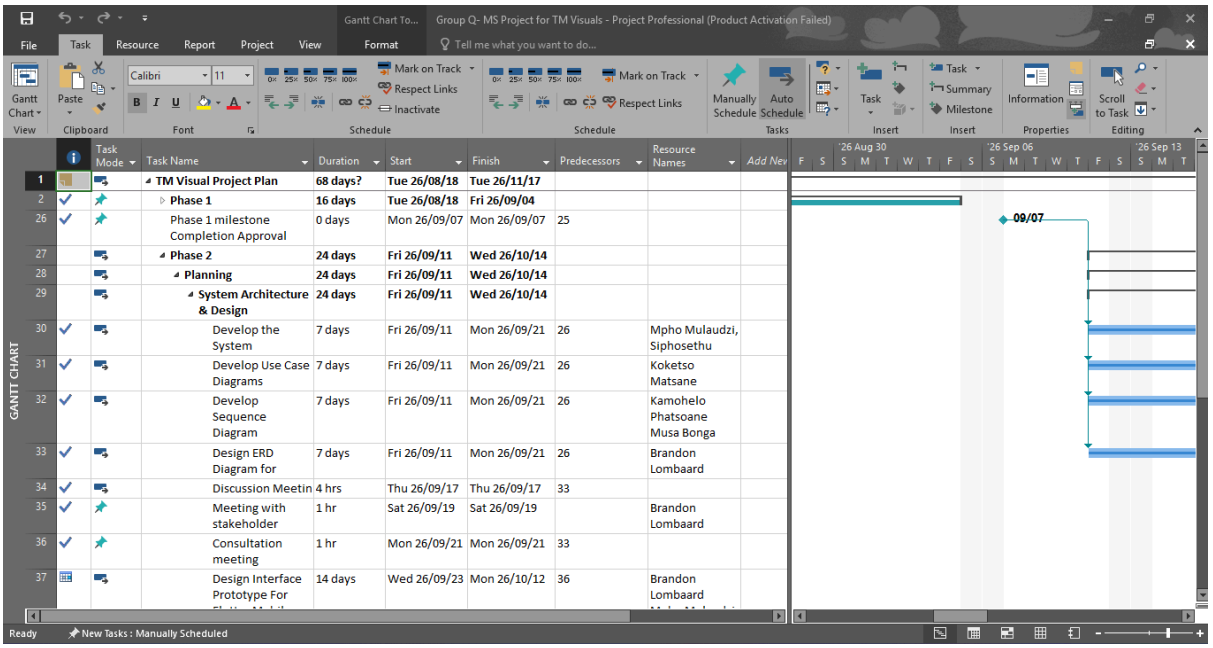
19.6 Design Alignment Table

Requirement / Function	Use Case	Sequence Diagram Activity	ERD Elements	Interface Design

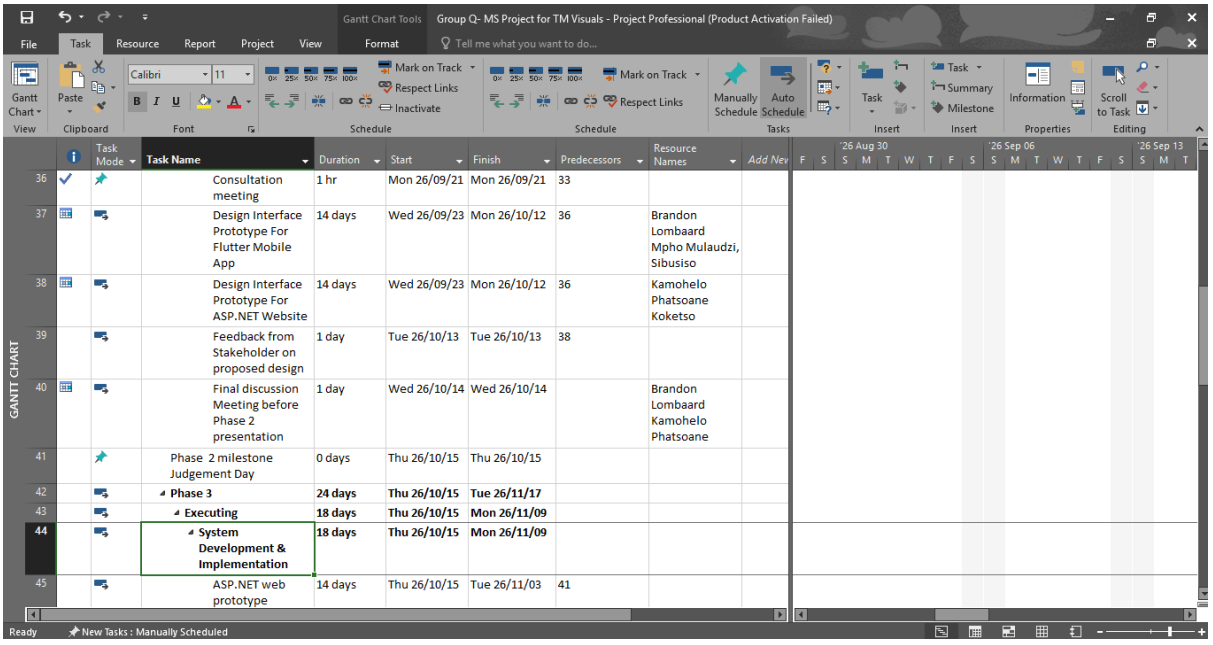
Users must register and log in securely.	Register, Login, Authenticate User	The user registers or logs in and the system authenticates the user.	USER, CLIENT, ADMIN	Sign Up, Login and Profile screens/pages
Visitors must browse services and portfolio work.	View Services, View Portfolio, Browse by Category	The visitor browses the website before registering or logging in.	SERVICE, PORTFOLIO_ITEM, PORTFOLIO_CATEGORY	Home, Services and Portfolio pages
Registered clients must check availability before booking.	Check Availability	The system retrieves and displays available time slots.	TIME_SLOT, BOOKING_SLOT	Availability screen/page
Registered clients must submit a booking request.	Make Booking	The client selects services and the system creates the booking record.	BOOKING, BOOKING_SERVICE, BOOKING_SLOT, SERVICE	Booking screen/page
Clients must receive booking-status updates.	View Booking, Booking Confirmation Notification	The system updates the booking status and sends a notification.	BOOKING, NOTIFICATION	My Bookings and Notifications screens/pages
Administrators must approve or reject bookings.	Manage Bookings, Approve Booking, Reject Booking, Update Booking Status	The administrator reviews the booking and updates its status.	ADMIN, BOOKING, NOTIFICATION	Admin Dashboard and Manage Bookings page

Clients must pay a deposit and receive an invoice.	Make Booking, View Booking	The deposit is recorded, payment is confirmed and an invoice is issued.	DEPOSIT_RECORD, INVOICE, BOOKING	Invoice and Deposit Management page; My Bookings screen
Administrators must manage services and portfolio content.	Manage Services, Manage Portfolio, Categorise Portfolio Content	The administrator adds, edits or removes service and portfolio information.	SERVICE, PORTFOLIO_ITEM, PORTFOLIO_CATEGORY, ADMIN	Manage Services and Manage Portfolio pages
Administrators must upload completed client work.	Upload and Manage Completed Content	The administrator uploads content and provides a delivery link.	CLIENT_CONTENT, BOOKING, ADMIN	Content Upload page
Clients must access and download completed work.	Access Private Client Content, Download Content	The client logs in and the system retrieves the authorised content link.	CLIENT_CONTENT, BOOKING, CLIENT	Private Content and My Gallery screens/pages

20. MS Project Current progress:



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1		**TM Visual Project Plan**	**68 days?**	**Tue 26/08/18**	**Tue 26/11/17**		
2		**Phase 1**	**16 days**	**Tue 26/08/18**	**Fri 26/09/04**		
26		Phase 1 milestone Completion Approval	0 days	Mon 26/09/07	Mon 26/09/07	25	
27		**Phase 2**	**24 days**	**Fri 26/09/11**	**Wed 26/10/14**		
28		**Planning**	**24 days**	**Fri 26/09/11**	**Wed 26/10/14**		
29		**System Architecture & Design**	**24 days**	**Fri 26/09/11**	**Wed 26/10/14**		
30		Develop the System	7 days	Fri 26/09/11	Mon 26/09/21	26	Mpho Mulaudzi, Siphosethu
31		Develop Use Case Diagrams	7 days	Fri 26/09/11	Mon 26/09/21	26	Koketso Matsane
32		Develop Sequence Diagram	7 days	Fri 26/09/11	Mon 26/09/21	26	Kamohelo Phatsoane Musa Bonga
33		Design ERD Diagram for	7 days	Fri 26/09/11	Mon 26/09/21	26	Brandon Lombaard
34		Discussion Meetin	4 hrs	Thu 26/09/17	Thu 26/09/17	33	
35		Meeting with stakeholder	1 hr	Sat 26/09/19	Sat 26/09/19		Brandon Lombaard
36		Consultation meeting	1 hr	Mon 26/09/21	Mon 26/09/21	33	
37		Design Interface Prototype For	14 days	Wed 26/09/23	Mon 26/10/12	36	Brandon Lombaard



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
36		Consultation meeting	1 hr	Mon 26/09/21	Mon 26/09/21	33	
37		Design Interface Prototype For Flutter Mobile App	14 days	Wed 26/09/23	Mon 26/10/12	36	Brandon Lombaard Mpho Mulaudzi, Sibusiso
38		Design Interface Prototype For ASP.NET Website	14 days	Wed 26/09/23	Mon 26/10/12	36	Kamohelo Phatsoane Koketso
39		Feedback from Stakeholder on proposed design	1 day	Tue 26/10/13	Tue 26/10/13	38	
40		Final discussion Meeting before Phase 2 presentation	1 day	Wed 26/10/14	Wed 26/10/14		Brandon Lombaard Kamohelo Phatsoane
41		Phase 2 milestone Judgement Day	0 days	Thu 26/10/15	Thu 26/10/15		
42		**Phase 3**	**24 days**	**Thu 26/10/15**	**Tue 26/11/17**		
43		**Executing**	**18 days**	**Thu 26/10/15**	**Mon 26/11/09**		
44		**System Development & Implementation**	**18 days**	**Thu 26/10/15**	**Mon 26/11/09**		
45		ASP.NET web prototype	14 days	Thu 26/10/15	Tue 26/11/03	41	

21. Stakeholder Feedback

For the purposes of this document, the stakeholder will be referred to as TM.

Overall Opinion on the Design

TM expressed satisfaction with the overall design and layout of the system. He particularly appreciated its simplicity, straightforward functionality, and ease of navigation.

Questions and Concerns

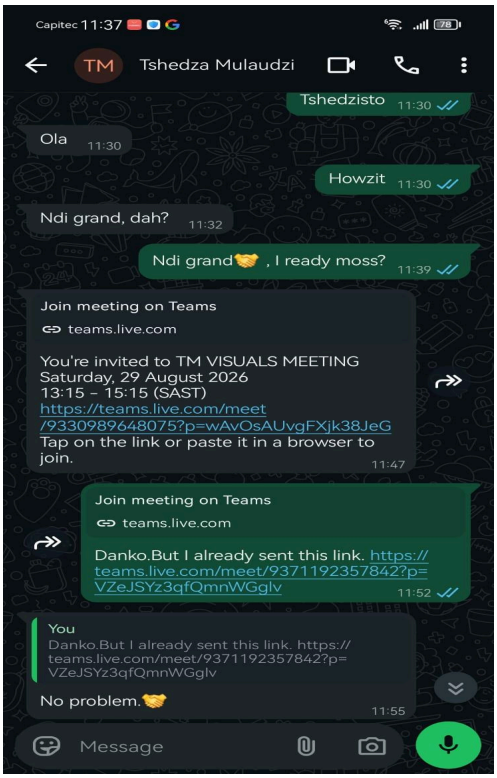
TM raised questions regarding the system's security and user access management. In particular, he wanted clarification on whether clients would be able to access the system's administrative functionality and how the system would distinguish between administrator and standard user accounts.

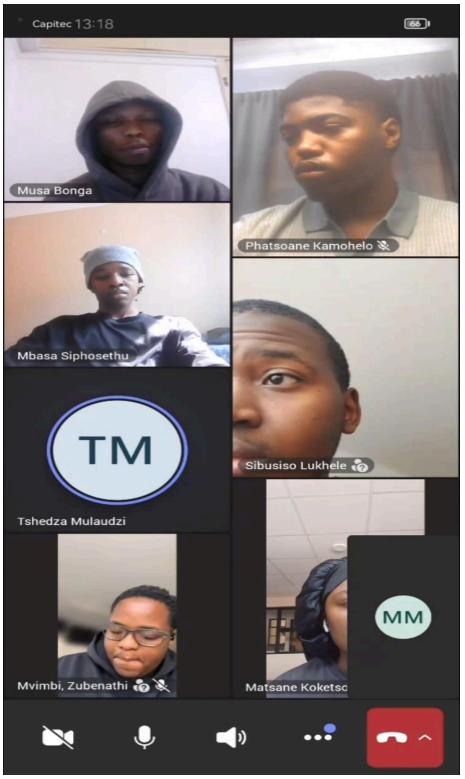
Feedback and Feature Suggestions

TM suggested introducing a feature that would allow clients to specify their intended budget when booking services. This information could assist TM during price negotiations and help him better accommodate clients' financial expectations when finalising service prices.

Recommended Action

The implementation of the proposed budget feature will be discussed and agreed upon by all team members, taking into consideration the limited time available for development.





Communication Artifacts:

- WhatsApp
- Teams

Meeting Notes:

STAKEHOLDER INTERVIEW AND PROCESS UNDERSTANDING

Stakeholder Profile

Stakeholder: Tshedza Mulaudzi
Business: TM Visuals
Role: Owner and primary operator
Business Type: Photography and Videography

1. Objectives of the Stakeholder Interview

The purpose of the stakeholder interview was to:

- Understand how TM Visuals currently operates.
- Understand the business's daily processes.
- Identify problems and challenges experienced by the business.
- Understand the different types of customers served by the business.
- Understand how customers currently interact with TM Visuals.
- Understand the current booking and client management process.
- Identify how photography and videography content is managed and delivered.
- Understand the stakeholder's expectations and needs.

2. Interview Preparation and Recording

During the interview preparation, the following practices were identified:

- Multiple team members should record the interview.
- Multiple recordings provide backups if one recording device fails.
- Team members should take notes while the interview is taking place.
- Manual notes help capture important details that may not be clearly captured in the
- recording.
- The team should consider the storage capacity of recording devices.
- Devices with sufficient storage should be used when recording longer sessions.
- Cameras may be used so that observers, such as lecturers, can identify the team
- members speaking.
- The stakeholder's privacy and confidentiality should be respected when recording and
- presenting the interview.

3. Business Services

- TM Visuals is a creative service business specialising in visual media.
- The main services provided are:
- Photography.
- Professional photo shoots.
- Videography.
- Music video production.
- Other video content.

4. Customers Served by TM Visuals

The business serves different types of customers, including:

- Individuals who require photography or videography services.
- Students
- Students requiring photography shoots.
- Shoots may take place on campus.
- Event Organisers
- Professionals who organise different types of events.
- They may require photography or videography services.
- Restaurants
- Restaurants requiring food photography.
- Restaurants may also require commercial photography.

5. Current Communication Channels

- TM Visuals currently relies on social media and instant messaging to communicate with customers and promote the business.

The main platforms used are:

Instagram

- Used to post previous work.
- Used to interact with customers.

TikTok

- Used to display content.
- Used to reach potential customers.

WhatsApp

- Used as the primary method of direct communication with clients.
- Used to discuss bookings.
- Used to communicate with customers about their requirements.
- The business currently does not have a centralised website or application.

6. Current Booking Process

The current booking process is handled mainly through direct communication with the client.

The process begins when:

A client contacts the photographer after hearing about the business.
The client may contact the photographer through WhatsApp or another messaging platform.

The photographer then asks the client for information such as:

- Type of event.
- Date.
- Location.
- Duration of coverage.
- Expected number of attendees.
- Other requirements related to the shoot.

7. Pricing Process

After gathering the required information:

- The photographer provides the client with a quotation.
- The price may depend on the requirements of the shoot.
- The price can be negotiated with the client.
- The client's budget may be considered.
- The final price is agreed upon between the photographer and the client.

For example:

- A two-hour shoot may initially be quoted at R2,000.
- The final price may change depending on the client's requirements and budget.
- The stakeholder explained that pricing should not be rigidly based on the type of event.
  For example:
  A wedding in Limpopo may have a different price from a wedding in Sandton.
  Location and travel requirements can affect the final price.

8. Deposits and Booking Confirmation

The stakeholder explained that:

- A booking is not officially confirmed until a deposit has been paid.
- The standard deposit is 50% of the total agreed amount.
- The deposit is non-refundable.
- The non-refundable deposit policy must be communicated to the client beforehand.

The deposit is used to:

- Confirm that the client is serious about the booking.
- Provide financial security to the photographer.
- Protect the photographer from losses caused by cancellations.

9. Reason for the Non-Refundable Deposit

The stakeholder explained that the deposit is non-refundable because:

Once a client confirms a booking, the photographer reserves the date. Other clients may be turned away for the same date. If the client cancels later, the photographer may not be able to find another client for that date and this can result in a loss of income. The deposit helps compensate for the lost opportunity.

10. Transportation Costs

Transportation costs are considered when a client is located outside the photographer's immediate area.

The photographer should:

- Determine the client's location.
- Consider the distance that needs to be travelled.
- Calculate the transportation costs.
- Include transportation costs in the total charge where necessary.

For example:

If the photographer is based in Johannesburg and is booked for a shoot in Pretoria, an additional transportation fee may be charged. The fee would help cover the round-trip travel costs.

11.Invoice Numbering and Record-Keeping

- The stakeholder discussed the use of sequential invoice numbers to keep track of
- business activities.
- Invoice numbering can help the photographer:
- Keep a chronological record of events.
- Identify the date of an event.
- Identify the type of event.
- Track the total number of events completed within a month.
- Monitor business activity from the beginning of the business.
- Maintain an organised record of completed work.
- These records also assist with scheduling shoots.

12.Client Acquisition

The stakeholder explained that clients are acquired through different methods.

Important methods are:

- Word of mouth

- Existing clients recommend TM Visuals to other people.
- New clients may contact the photographer directly through WhatsApp.
- A challenge occurs when:
- A person receives a recommendation but does not use Instagram or TikTok.
- The person cannot easily view the photographer's existing portfolio.
- The photographer may then need to send previous work directly to the potential client.

13.Portfolio and Privacy Concerns

The stakeholder explained that showing previous work is important because clients generally want to see the quality of a photographer's work before booking.
However, sending previous work directly to individuals creates privacy concerns.

The concerns include:

- The photographer cannot fully control what happens to the images after sending them.
- The recipient may save the images.
- The recipient may redistribute the images.
- The photographer may not have control over how the images are used.
- Despite these concerns, showing previous work remains important when trying to
- secure a new client.

14.Website Purpose

The stakeholder explained that the purpose of having a website would be to provide a centralised and professional presence for TM Visuals.

The website would allow the business to:

- Introducing the photographer.
- Provide background information about the business.
- Explain the photographer's experience.
- Show the types of events handled.
- Display the business location.
- Showcase previous work.
- Provide access to social media platforms.
- Make it easier for potential clients to view the photographer's work.
- The website would particularly help clients who hear about TM Visuals through word of
- mouth but do not use social media.

15.Portfolio Categories

The stakeholder indicated that the portfolio should be organised into different categories.

The categories discussed include:

- Food
- Food photography and videography.
- Culinary content.
- Graduation
- Graduation photography.
- Academic ceremony coverage.
- Weddings and Parties
- Wedding photography.
- Wedding videography.
- Party and event coverage.
- Branding
- Commercial content.
- Content created for brands.

16.Users of the Business

The stakeholder identified three main groups that would interact with the system:

- Potential Clients
- Students.
- Individuals.
- Event organisers.
- Brands.
- Existing Clients
- Clients who have already used TM Visuals.
- Clients who need to access their completed work.
- Business Owner
- The owner of TM Visuals.
- Responsible for monitoring and managing the business.

17.Content Delivery

The stakeholder identified content delivery as an important part of the photography and videography process.

The photographer currently uses:

- Google Drive

- Completed work is uploaded to Google Drive.
- A link is then sent to the client.
- This allows the client to receive high-quality files.
- The files can remain accessible for a longer period.

The stakeholder explained that:

- WhatsApp is convenient for communication.
- However, WhatsApp can reduce the quality of images and videos.
- Google Drive is therefore preferred for delivering completed high-quality work.

18.Content Organisation

The stakeholder explained that completed jobs are organised using specific information.

The information used includes:

- Client name.
- Date of the shoot.
- Type of content.
- Type of event.

For example:

- A client’s name can be used to identify their work.
- The date can be used to identify when the work was completed.
- The content type can identify whether the work was a music video, event, or another
- type of shoot.
- This method helps the photographer locate work quickly when multiple shoots have
- been completed within a short period.

19.Access to Client Content

The stakeholder explained that client content is shared using links.

The process includes:

- Creating a folder containing the client’s work.
- Providing the client with a unique link.
- Controlling who can access the folder.
- Deciding whether the link can be accessed by anyone or requires specific approval.
- The stakeholder also made a distinction between:
- Public content intended to showcase the business.
- Private client content intended only for the specific client.

20.Public and Private Content

The stakeholder explained that not all content should be accessible to everyone.

Public Content

- Used to showcase the photographer's work.
- Can be viewed by potential clients.
- Intended for portfolio purposes.

Private Client Content

- Contains photographs or videos belonging to a specific client or event.
- Should only be accessible to the relevant client.
- Requires controlled access.

21.Booking Availability

The stakeholder explained that availability is important when managing bookings.

The business needs to:

- Know which dates are already booked.
- Avoid accepting multiple bookings for the same date.
- Keep track of confirmed bookings.
- Ensure that a date is no longer available once a booking has been confirmed through
- payment of the required deposit.

22.Pricing Policy

The stakeholder explained that TM Visuals does not use fixed prices for every type of event.

Instead:

- Prices are negotiated with clients.
- The final price depends on the circumstances of the booking.
- The client's requirements are considered.
- Location can affect the price.
- Travel costs can affect the price.
- The scope of the work can affect the price.
- The client's budget can be considered.
- The stakeholder emphasised that pricing should be based on an agreement between
- the photographer and the client rather than a rigid event category.

23.Creative Tools Used

The stakeholder explained that several software platforms are used to produce and edit Content.

These include:

Adobe Photoshop

Used for photo editing.

CapCut

Used for video editing.

TikTok

Used for video content and social media exposure.

Google Drive

Used for storing and delivering completed work.

24.Relationship Between the Business Owner and Developers

The stakeholder emphasised that the relationship between the business owner and developers should be based on:

- Teamwork.
- Communication.
- Cooperation.
- Continuous interaction.
- Problem solving.

The stakeholder explained that, as the business owner is not a web developer, there should be a way to communicate technical problems to the development team.

Examples of possible problems include:

- Website crashes.
- Client access problems.
- Technical errors.
- Problems accessing content.
- Other website-related issues.

The development team should assist with technical problems that the business owner cannot resolve independently.

Agenda of the stakeholder meeting:

Kick-Off Meeting

29 August 2026

Time: 13h00

Location: Microsoft Teams

Project Name: Photographer and client management Project

Meeting Objective: Get the project off to an effective start by introducing the stakeholder, review project goals, and discuss future plans.

Attendees: Mulaudzi Tshedza, Phatsoane KTJ, Lukhele S, Matsane K, Bonga M, Mbasa S, Mulaudzi MF, and Mvimbi ZSZ.

Agenda:

- **Review of the Project Background**
- **Discussion of project organizational structure**
- **Elicit stakeholder expectations**
- **Discussion of scope goals**
- **Discussion of other important details**

Role Selection:

